Planning as Autoregressive Sequence Modelling: Decoding Strategies

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The problem of finding a sequence of actions that achieves a goal in a given environment is known as planning. This project investigates the planning problem of finding sequences of moves, called trajectories, from starting positions to goal positions in different grid-based mazes. Figure 1 shows an example of such a trajectory. A transformer is a type of neural network commonly used in language models to predict probabilities of sequences of words. This project, however, trains a Transformer to predict probabilities of sequences of actions. The rule that is used to choose the next move in a trajectory from these probabilities is known as the decoding strategy. The central research question for this project then is: how does the choice of decoding strategy affect the quality of the generated trajectories for maze planning?

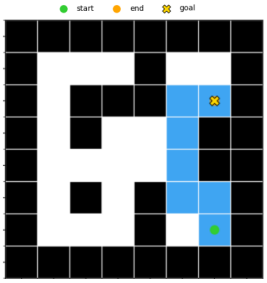


Figure 1: Example of a valid trajectory.

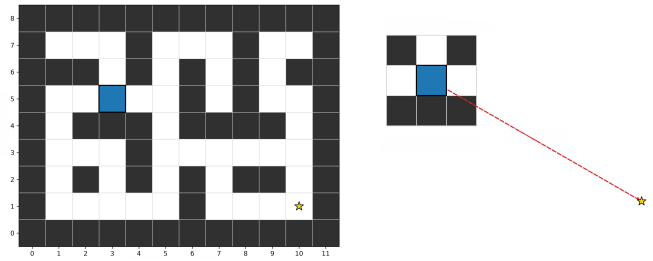


Figure 2: The complete maze state (left) and the information available to the model in the partially observed setting (right).

A series of experiments was conducted to compare three decoding strategies: greedy decoding, beam search with width $B = 2$, and beam search with width $B = 3$. They were compared in terms of how often they reached the goal, how many steps were taken before completion and how much computation time they required. Greedy decoding always chooses the move with the highest predicted probability. Beam search keeps track of the B most probable sequences of moves at each step, extending each by one move and choosing the most probable out of the B retained sequences at the end [1].

The first experiment was conducted in maze environments where the model was given the entire maze layout and the start and goal positions. The model therefore knows its exact position in the maze at every step. Beam search reached the goal more often and took fewer moves than greedy decoding, with the wider beam performing best. Although each move took slightly longer to compute, beam search required less computation time per planning attempt overall.

The second experiment compared the same decoding strategies using a larger training dataset. In this setting, the model was not given the complete maze layout or the coordinates of its position and the goal. Instead, at each position, it received a 3×3 view of the surrounding squares, the goal's position relative to its current position, and the previous observations and moves. At every position in the maze, the model only "sees" information of the kind shown in Figure 2. In this setting, beam search reached the goal more often and in fewer moves than greedy decoding, with the wider beam performing slightly better. Greedy decoding, however, was much faster both per move and per planning attempt.

[1] Daniel Jurafsky and James H. Martin. *Speech and Language Processing*, 3rd edition, online manuscript, 2026.